Advancing Machine Olfaction for Real-World Allergen Detection

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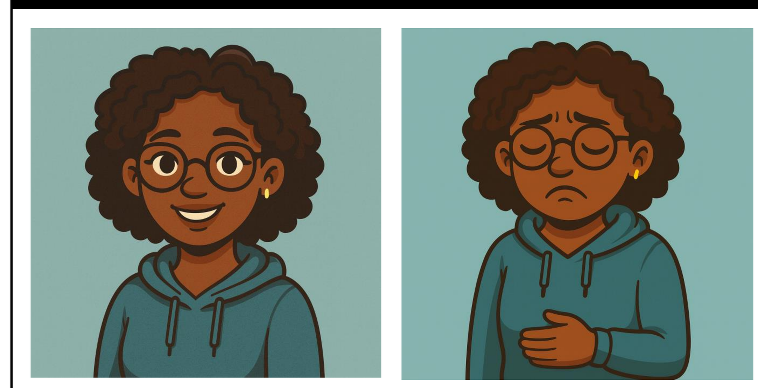
ABSTRACT

Machine olfaction—training AI to recognize smells—is crucial for food allergen detection and environmental monitoring, but current datasets typically focus on pure substances rather than realistic mixtures. To address this, we extend SMELLNET, a large-scale benchmark for odor recognition using portable gas sensors, by curating a new dataset of peanut-containing food mixtures and developing specialized ML models for mixture classification. Evaluated under controlled conditions, our models show improved accuracy and robustness. Preliminary real-time inference confidently detects peanuts (~80% probability) while assigning lower scores (~42%) to non-peanut samples, highlighting strong potential for practical deployment. Our work provides new data, models, and benchmarks, advancing machine olfaction toward scalable real-world applications.

INTRODUCTION

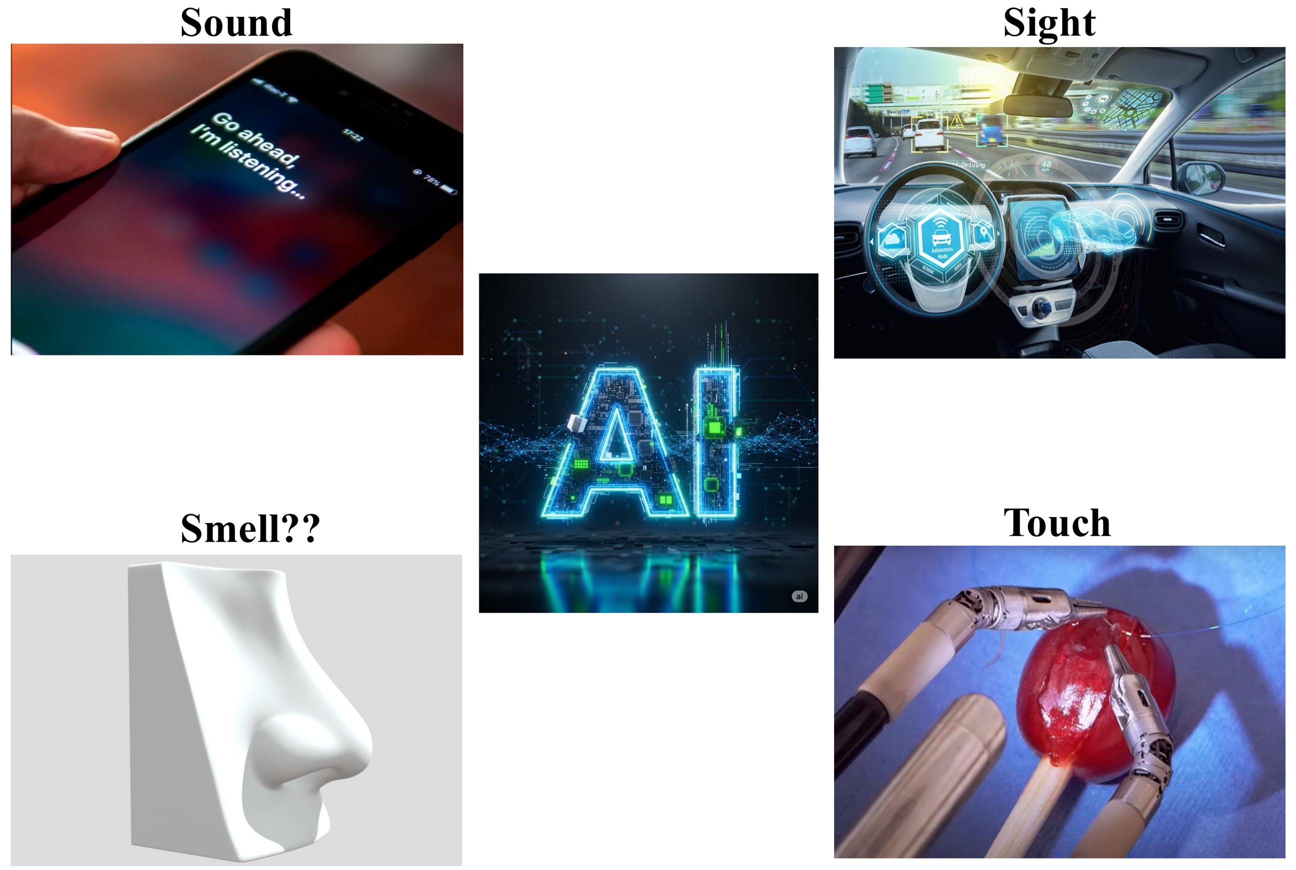


IDEAL WORLD REALITY

Research shows that **1 in 10** U.S. adults and **1 in 17** U.S. children have a food allergy [2][3].

AI has harnessed different modalities:

Sound Sight Smell?? Touch

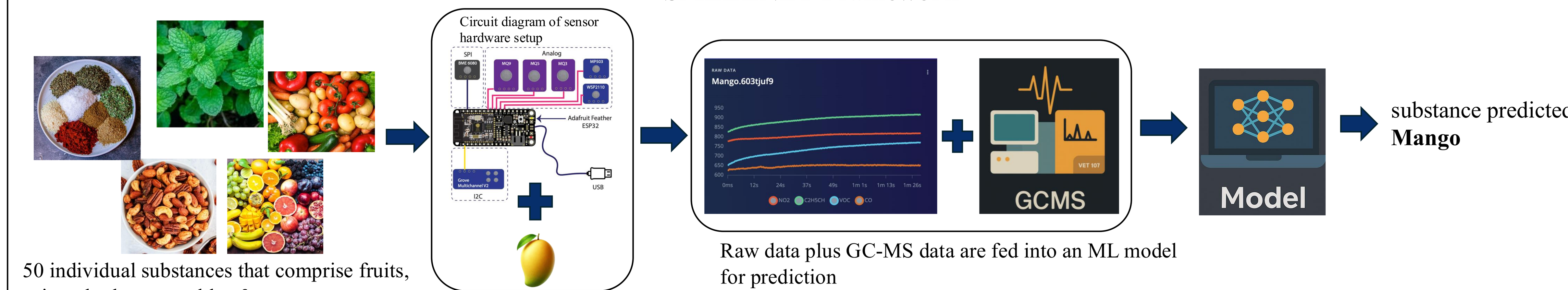


Why AI for smell is challenging: To handle smell properly, AI must learn rich temporal features, generalize from limited data, and cope with noisy environmental features.

Research Question: Can machine learning models trained on gas sensor data reliably detect peanut presence in mixed food samples like salads or chocolate?

METHODOLOGY

SMELLNET Framework

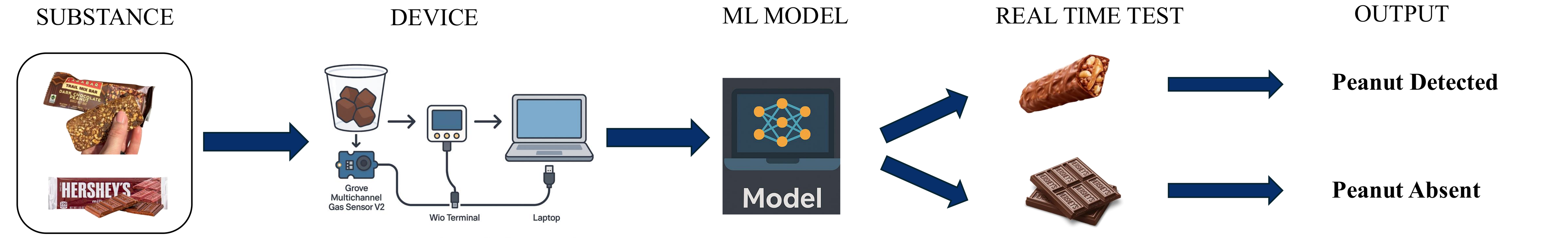


50 individual substances that comprise fruits, spices, herbs, vegetables & nuts.

Gas sensor collects odor data from a substance

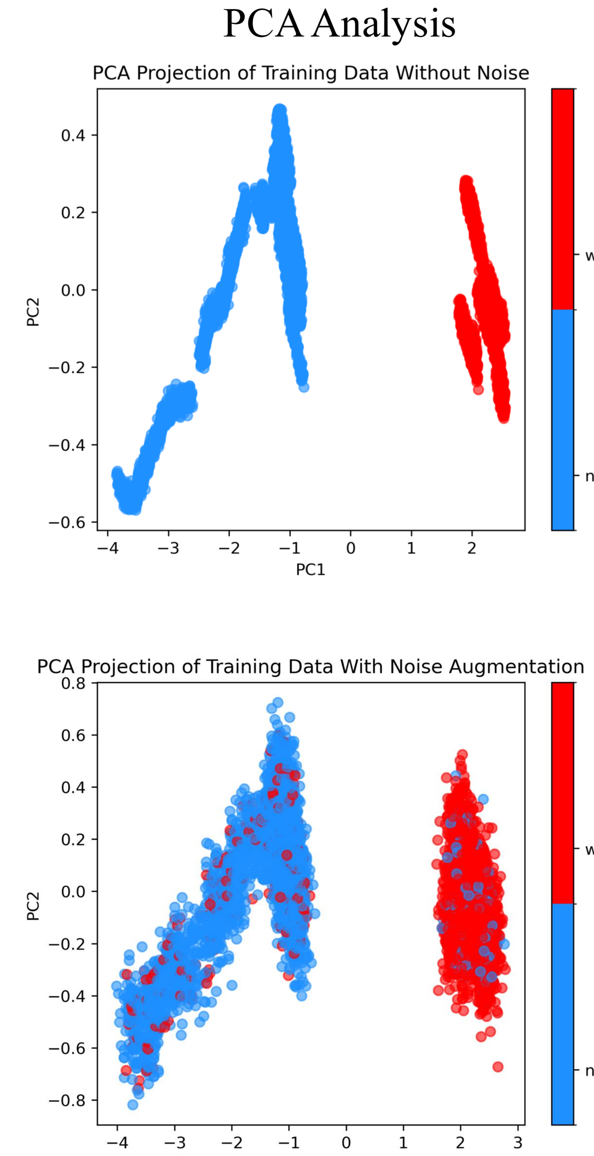
Raw data plus GC-MS data are fed into an ML model for prediction

Allergen Detection Pipeline

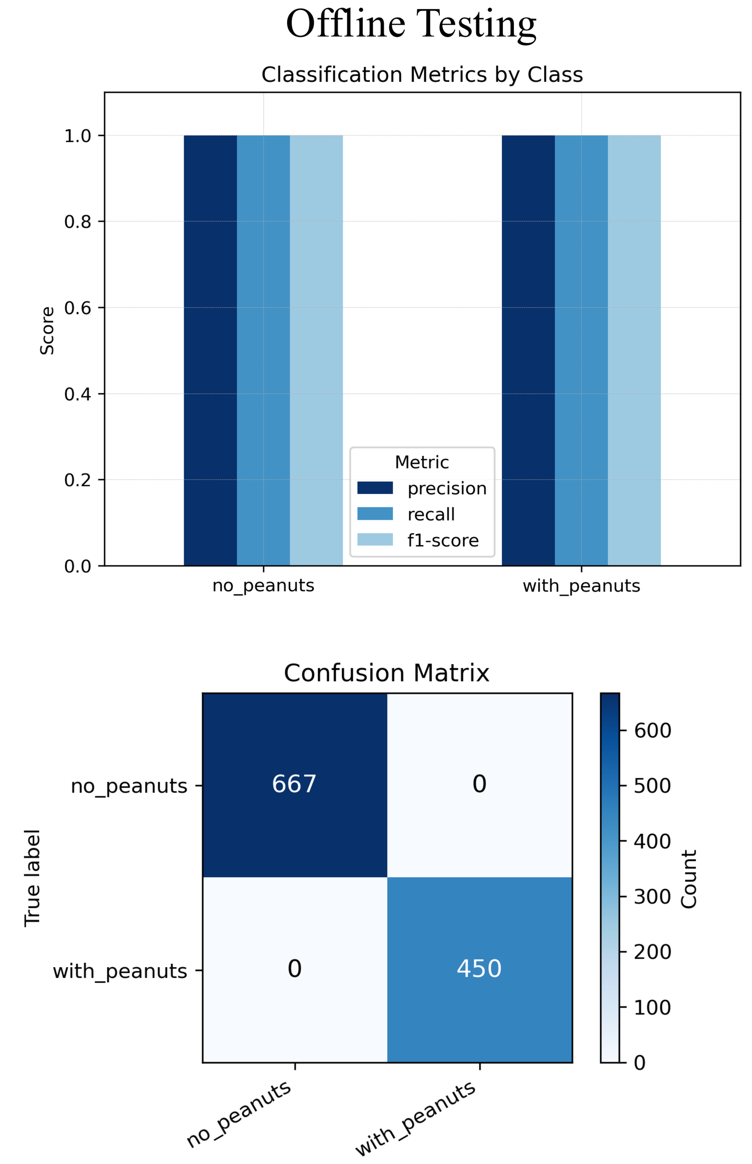


RESULTS

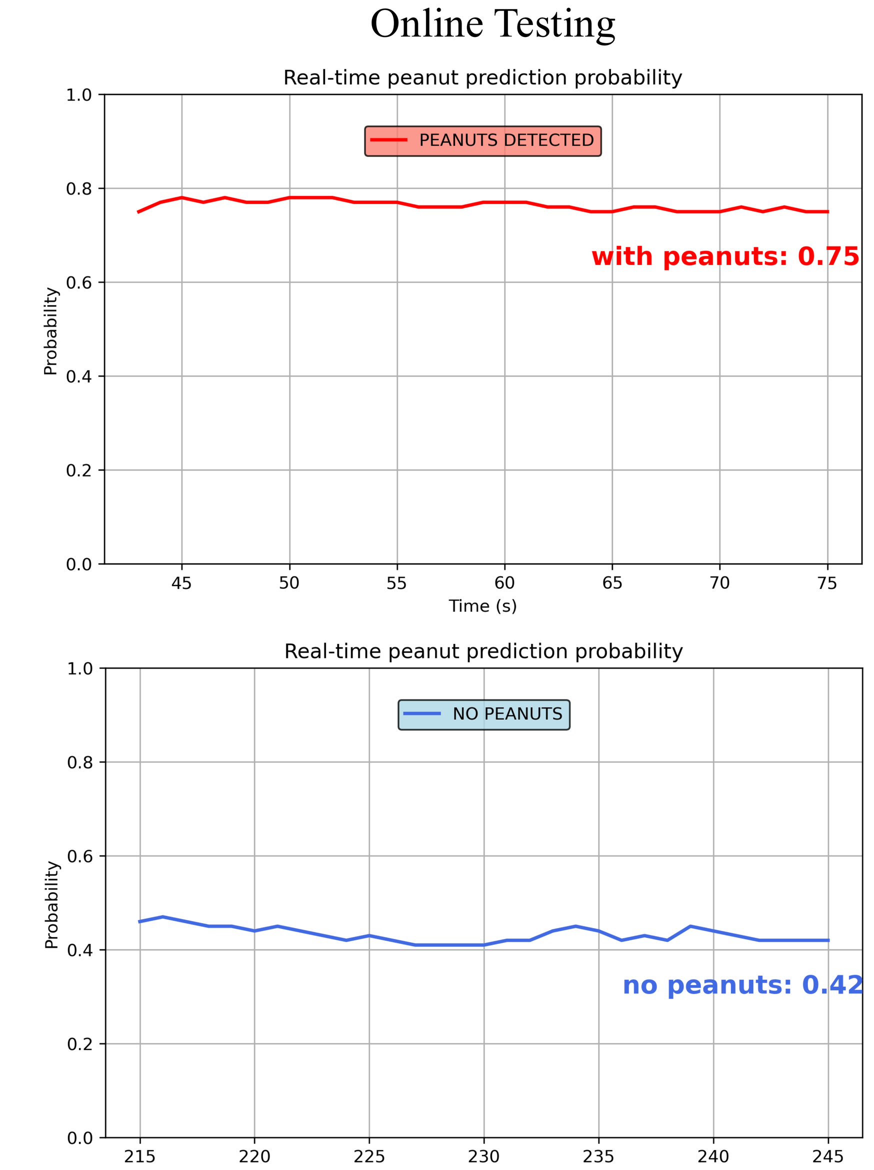
PCA Analysis



Offline Testing



Online Testing



CONCLUSION

- ❑ Principal Component Analysis (PCA) projects high-dimensional sensor data into two dimensions, separates the classes, and shows how added noise broadens clusters to improve model robustness.
- ❑ In a real-time prediction run, the model can confidently detect peanuts in a mixture (minimum of 70% probability) while maintaining below-threshold scores on a non-peanut sample (~42%).
- ❑ Preliminary real-time inference results show the model has strong potential for future deployment.

FUTURE WORK

- ❑ Collect more diverse data across different environments (e.g., open air, indoor, varied humidity/temperature) to improve model generalization.
- ❑ Train new models using time series and contrastive learning approaches, incorporating more real-world peanut mixture data.

POSSIBLE APPLICATIONS

- ❑ **SmellAI for food and retail**: Imagine sending a photo of your meal to a friend, and they can actually smell it through a device—this enables immersive food-sharing and enhances virtual retail experiences.
- ❑ **SmellAI for education**: Integrating scent with learning materials can aid memory recall and engagement, especially in fields like biology, history, or culinary education.
- ❑ **SmellAI for mental health**: When stress indicators are detected via sensors, the system can automatically release calming scents to help regulate mood and support mental well-being.

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